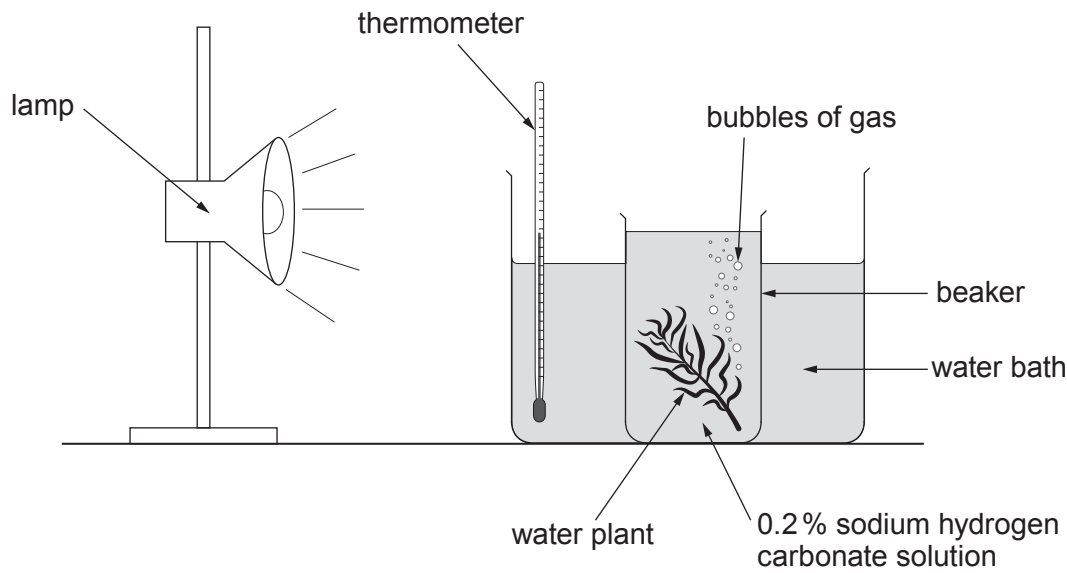


8. Students were instructed to investigate the effect of temperature on photosynthesis. They used the apparatus shown in **Image 8.1**.

Image 8.1



- The number of bubbles of gas produced by the water plant per minute was counted at different temperatures.
- The experiment was carried out three times at each temperature.
- The results are shown in **Table 8.2**. Means were calculated to the nearest whole number.

Table 8.2

Temperature of water bath (°C)	Number of bubbles of gas produced per minute			
	Test 1	Test 2	Test 3	Mean
20	18	21	12	17
25	15	24	21	20
30	24	27	30	27
35	26	25	26	26
40	14	13	10	12
45	3	5	6	
50	0	0	0	0



Examiner
only

(a) (i) State the name of the gas produced during photosynthesis. [1]

.....

(ii) State **two** ways in which plants use the glucose produced during photosynthesis. [2]

I.

II.

(b) (i) Calculate the mean number of bubbles per minute for 45 °C. **Write your answer in Table 8.2.** [2]

(ii) Describe the relationship between the temperature and the number of bubbles produced per minute. [2]

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(iii) Explain the result for 50 °C. [2]

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9

